

Learning Objective 2.6: I can measure the impedance and S-parameters of an antenna using a vector network analyzer and interpret the results.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Four names for one number.** A one-port measurement gets reported four different ways depending on which menu you open. Reference impedance is $Z_0 = 50 \Omega$ throughout.

- (a) An antenna reads $|S_{11}| = -14 \text{ dB}$ at its design frequency. Find $|\Gamma|$, the VSWR, and the percentage of incident power reflected.

$$|\Gamma| =$$

$$0.199$$

$$$$

- (d) Lesson 7 predicted $Z_{\text{in}} = 73 + j42.5 \, \Omega$ for an ideal half-wave dipole. Compute $|\Gamma|$, $|S_{11}|$ in dB, and the VSWR. Does the textbook dipole meet the -10 dB specification, and what one change fixes it?

$$\begin{aligned}\Gamma &= \frac{23 + j42.5}{123 + j42.5} \\ |\Gamma| &= \frac{48.3}{130.1} = 0.371 \\ |S_{11}| &= 20 \log_{10}(0.371) = -8.6 \text{ dB} \\ \text{VSWR} &= \frac{1.371}{0.629} = 2.18\end{aligned}$$

$|S_{11}| =$

-8.6 dB

$\text{VSWR} =</$

2. **From a marker readout to an impedance.** A calibrated VNA marker sitting at 1.575 GHz on a small GPS-band element reads $S_{11} = 0.33 \angle 155^\circ$.

- (a) Write Γ in rectangular form and find the input impedance Z_{in} .

$$\Gamma = 0.33 (\cos 155^\circ + j \sin 155^\circ) = -0.299 + j0.140$$

$$Z_{\text{in}} = Z_0 \frac{1 + \Gamma}{1 - \Gamma} = 50 \frac{0.701 + j0.140}{1.299 - j0.140}$$

$$Z_{\text{in}} = \boxed{26.1 + j8.2 \, \Omega}$$

Multiplying numerator and denominator by the conjugate of the denominator,

3. **Bandwidth off a sweep.** A calibrated, load-verified sweep of a wire dipole produced the following magnitudes.

f (MHz)	850	870	880	890	900	905	910	920	930	950
$ S_{11} $ (dB)	-3.8	-6.5	-8.8	-12.4	-17.7	-19.4	-17.8	-12.5	-9.0	-5.2

- (a) Identify the resonant frequency and say how you know.

The deepest point of the sweep is -19.4 dB at 905 MHz, and the trace is symmetric about it. That is the resonance: the reactance passes through zero there, so the reflection is smallest

4. **Calibration and the reference plane.** A classmate calibrates short–open–load at the end of the test cable, then screws a 10 cm RG-58 pigtail ($v_f = 0.66$) between that connector and the antenna before sweeping.

- (a) Name the three error terms a one-port SOL calibration removes, one sentence each, and explain why exactly three standards are required.

Directivity: the coupler that should sample only the returning wave leaks some of the outgoing wave into the same receiver, so the VNA sees a reflection even from a perfect load. **Source match:** the test port is not exactly $50\ \Omega$, so energy returned by the antenna is partly re-reflected back at it. **Tracking:** the reference and test receiver paths have different gain and phase, and both drift with frequency. Three unknowns require three independent known loads, which is exactly what short ($\Gamma = -1$), open ($\Gamma = +1$), and load ($\Gamma = 0$) provide.

- (b

5. **What the VNA cannot see.** The instrument reports one complex number per frequency. This problem is about the questions that number cannot answer.

- (a) You solder a $50\ \Omega$ chip resistor across the connector and sweep it. Predict $|S_{11}|$, the VSWR, and the radiated power. What does the result prove about what S_{11} measures?

$\Gamma = 0$, so $|S_{11}|$ drops to the noise floor of the instrument and the VSWR reads 1.00 at every frequency: a perfect match. It radiates **nothing** – all of the delivered power becomes heat. S_{11} therefore measures **mismatch only**. It cannot separate power that left as radiation from power that died as loss, so it can never confirm efficiency.

- (b) Antenna A measures $-25\ \text{dB}$ with $2.0\ \text{dBi}$ of gain; antenna B measures $-9\ \text{dB}$ with $5.0\ \text{dBi}$. Find the mismatch loss of each and decide which one puts more power